

Viva Voce Questions



1. What type of language is Python?

Ans. Python is a scripting language.

2. How is Python an interpreted language?

Ans. Python is an interpreted language because it executes code line by line at runtime without compiling into machine level code beforehand.

3. What are Python modules?

Ans. Python modules are files containing Python code. This code can either be functions, classes, or variables. A Python module is a .py file containing executable code.

4. Name some commonly used built-in modules in Python.

Ans. Some of the commonly used built-in modules are os, sys, math, random, datetime, JSON.

5. Is Python case-sensitive?

Ans. Yes, Python is a case-sensitive language.

6. Is indentation required in Python?

Ans. Indentation is necessary for Python. It specifies a block of code. All codes within loops, classes, functions, etc. are specified within an indented block. This is usually done using four space characters. If your code is not indented necessarily, it will not execute accurately and will throw errors as well.

7. How do you write comments in Python?

Ans. Comments in Python start with a # character. However, alternatively at times, commenting is done using docstrings (strings enclosed within triple quotes)

8. What is the purpose of is, not, and in operators?

Ans. Operators are specific special functions. They take one or more values and produce a corresponding result.

- a. is: returns true when two operands are true. (Example, "a" is 'a').
- b. not: returns the inverse of the boolean value.
- c. in: checks if some element is present in some sequence.

9.What does len () do?

Ans. It is used to determine the length of string, a list, an array, etc.

10.Is Python an interpreted language?

Ans. Yes, Python is an interpreted language.

11.Is Python object-oriented language?

Ans. Yes, Python is an object-oriented language.

12. How many modes are available in Python?

Ans. Interactive mode and script mode.

13. Define Python.

Ans. Python is a high-level, interpreted, interactive, and object-oriented scripting language.

14.Who developed Python?

Ans. Python was developed by Guido van Rossum in the late 80s and early 90s at the National Research Institute for Mathematics and Computer Science in the Netherlands.

15.Why Python is easy to learn?

Ans. Python has a relatively few keywords, simple structure, and clearly defined syntax. This allows the students to pick up the language in a relatively short period of time.

16.Explain the disadvantages of Python.

Ans. Disadvantages of Python are:

- a) Python is an interpreted language and is slow compared to C, C++, or Java.
- b) Python is not the best for memory-intensive tasks.

17. Can python be easily integrated with C++?

Ans. Yes, Python can be easily integrated with C++.

18.What is another name of comparison operator?

Ans. Relational operator.

19. Which operator is used for assigning a value to a variable?

Ans. Assignment operator.

20. Which operator is used for comparison of values?

Ans. Comparison operators.

21. What is the use of the "+" operator?

Ans. The "+" operator adds values on either side of the operator.

22. What is the use of the "*" operator?

Ans. The "*" operator multiplies values on either side of the operator.

23. What is the use of the "%" operator?

Ans. The "%" operator divides the left-hand operand by the right-hand operand and returns the remainder.

24. What is the use of the "**" operator?

Ans. The "**" operator performs exponential (power) calculation on operands.

25. What is the use of the "==" operator?

Ans. It checks if the value of two operands is equal or not. It returns True if the values are equal, otherwise false

26. What is the use of the "!=" operator?

Ans. It checks if the value of two operands is equal or not. If the values are not equal, then condition becomes true.

27. Why do you use the "<>" operator?

Ans. The operator historically was used in some programming languages, including older version of Python, to denote inequality. However, in Python, the preferred and standard operator for inequality is "!=" (not equal to).

28. Describe the use of the ">" operator.

Ans. It checks if the value of the left operand is greater than the value of the right operand. If yes, then condition becomes true.

29. Why you use the "<" operator?

Ans. The less than operator checks if the value of the left operand is less than the value of the right operand. If yes, then condition becomes true.

30. Write syntax for the greater than or equal to operator.

Ans. `a >= b`

31. Is "<" an assignment operator?

Ans. No, "<" is not an assignment operator as it is a relational operator.

32. Optimize these statements as a Python programmer.

- a. word 'word'
- b. print word.len()

Ans. Below is the answer

- a. word= "word"
- b. print len(word)

33. How many types of operations are supported by Python language?

Ans. Python language supports the following types of operations: arithmetic operations, assignment operations, bitwise operations, identity operations, comparison or relational operations, logical operations, membership operations.

34. Why do you use the break statement?

Ans. The break statement can be used to terminate the loop.

35. Write the syntax of an 'if' statement in Python programming language.

Ans. The syntax of an 'if' statement in Python programming language is

```
if (condition):  
    statement.
```

36. Write the syntax of an 'if..... else' statement in Python programming language.

Ans. The syntax of an 'if.....else' statement in Python programming language is

```
if (condition):  
    statement for True  
else:  
    statement for False.
```

37. Define while loop.

Ans. A while loop statement in Python programming language repeatedly executes a target statement as long as a given condition is true.

38. Write the syntax of a while loop.

Ans. The syntax of a while loop in Python programming is

```
while <condition>:
```

```
    statement.
```

39. What happens when the condition of the while loop becomes false?

Ans. When the condition becomes false, program control passes to the line immediately following the loop.

40. How can a programmer take user input in Python?

Ans. A programmer can take user input in Python using input () function.